

Research Objective

The main objective of this study is to examine whether natural disasters in Turkey lead to meaningful changes in internal migration patterns, with a specific focus on out-migration from affected provinces. The analysis covers the period from 2008 to 2024 and includes earthquakes, floods, and wildfires.

Beyond identifying whether migration changes occur, the study also aims to understand **how these changes differ by disaster severity** and whether migration responses are temporary or persistent. In addition, a machine learning component is included to explore whether post-disaster migration changes can be partially predicted using observable characteristics such as pre-event migration trends, disaster type, and location.

Data and Variables

The analysis relies on annual province-level internal migration data obtained from TÜİK (Turkish Statistical Institute). The dataset includes inflow, outflow, and net migration for each province. From these variables, the **out-migration growth rate** is calculated as a year-over-year percentage change. This variable is chosen as the main outcome because it reflects changes in migration momentum rather than absolute population size.

Disaster event data are compiled from official sources, including AFAD for earthquakes and publicly available reports for floods and wildfires. Each event is associated with a specific province and year. Earthquakes are further classified into **major-disruptive** and **minor/non-disruptive** events based on real-world impact rather than magnitude alone.

Methodology and Code Workflow

The analysis follows an event-based pre/post design, implemented step by step through Python code.

First, all datasets are loaded and cleaned. Province names are standardized using a custom normalization function to avoid inconsistencies caused by capitalization or Turkish characters. Migration data are structured as a panel dataset indexed by province and year.

Second, exploratory data analysis is conducted using event-centered time-series plots. For each disaster event, out-migration growth is plotted for several years before and after the event year. The event year itself is explicitly marked to visually inspect whether a structural break occurs. This step is crucial for understanding whether disasters coincide with noticeable changes in migration trends.

Third, a pre/post comparison is performed numerically. For each event, average migration outcomes in the pre-event window are compared to those in the post-event window. The difference between these averages is used to construct event-level outcome variables, such as the change in out-migration growth.

Finally, a machine learning framework is applied at the event level. Each disaster event is treated as one observation, with features including disaster type, earthquake severity, province identity, and pre-event migration indicators. Two models are used: Ridge Regression as a linear and interpretable baseline, and Random Forest as a non-linear model capable of capturing complex interactions. Due to the small number of events, Leave-One-Out Cross-Validation (LOOCV) is used for evaluation.

Results and Interpretation

The results show clear differences across disaster types and severity levels. Major-disruptive earthquakes are associated with strong deviations from pre-event migration trends, often resulting in sharp declines or instability in out-migration growth. These effects are not merely short-term shocks but appear to represent structural breaks in demographic behavior.

In contrast, minor earthquakes generally do not produce persistent changes. Migration trends tend to follow their pre-event trajectory, suggesting that historical momentum plays a dominant role when physical and economic disruption is limited.

Floods and wildfires display more heterogeneous and temporary effects. While short-term fluctuations are observed, these events rarely lead to sustained changes in migration patterns.

The machine learning results reinforce these findings. The Random Forest model outperforms the linear model and identifies **pre-event out-migration growth** as the most important predictor. This suggests that migration responses are largely shaped by existing trends, with disaster severity acting as a threshold that determines whether those trends are interrupted.

Limitations and Self-Assessment

As a student project, this study has several limitations relating statistical power and predictive accuracy. The use of annual data also prevents the analysis of very short-term displacement immediately following disasters.

I am aware that this work is not methodologically perfect. However, the goal is not to produce definitive policy predictions, but to demonstrate a structured and transparent

analytical approach using real-world data. The limitations themselves highlight important directions for future research.

Future Directions and Social Implications

This project points to several possible extensions. Collecting more detailed data at the district or neighborhood level would allow for a more precise analysis of displacement dynamics. Including neighboring provinces could also help capture spillover effects, where people move not far away but just outside the affected area.

More importantly, machine learning models trained on richer datasets could eventually be used in a forward-looking way. After a major disaster, such models could help estimate **which cities are likely to receive migrants and in what proportions**. This information could guide the direction, location, and scale of social support projects, housing initiatives, and disaster response planning.

From the perspective of search and rescue and disaster preparedness, such predictive tools could contribute to more adaptive and proactive social policies, rather than reactive responses after displacement has already occurred.

Conclusion

This study finds that disasters do not uniformly increase internal migration in Turkey. Instead, migration responses depend on disaster severity, pre-existing migration trends, and structural constraints faced by individuals. Major earthquakes emerge as threshold events capable of disrupting demographic momentum, while minor disasters often fail to overcome social and economic inertia.

By combining event-based analysis with machine learning, this project demonstrates how quantitative tools can be used to explore complex social responses to disasters. While limited in scope, the findings offer a foundation for future research and illustrate how data-driven approaches can inform both academic understanding and practical disas